

Prajwal Premkumar Khairnar

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WORK EXPERIENCE

AI Research Engineer, March 2026 – Present (3 months)

Arva AI (YC S24), London, U.K.

- First AI Research hire and #5 member of technical staff, leading company research strategy.
- Building and optimizing agentic systems for financial crime, compliance, and AML/KYC workflows.
- Owning AI evaluations, benchmarking, and vertical dataset development.
- Partnering cross-functionally with product engineers across AI and product initiatives.
- Solving novel and complex problems in financial crime compliance through applied AI research and experimentation.

Senior Data Scientist, Jan 2025 – March 2026 (1 year 3 months)

Data Scientist, Feb 2022 – Dec 2024 (2 years 11 months)

Northern Care Alliance NHS Foundation Trust, Manchester, U.K.

- Architected and shipped a trust-wide RAG-based chatbot serving 6,000+ users, building the retrieval pipeline with OpenAI APIs, organisation-specific grounding, and internal deployment as sole technical owner
- Researched and built a hybrid NLP and recommendation system for clinical coding using local SLMs, targeting approximately 5 minutes saved per patient encounter; lead author and presenter at the Health and Care Analytics Conference 2023
- Conducted NER research on clinical text using BioBERT and biomedical transformer models via HuggingFace, benchmarking fine-tuned domain-specific models against LLM-based extraction pipelines
- Designed and deployed a Did Not Attend risk prediction system with a custom internal web app surfacing real-time predictions to operational teams; accepted for presentation at HACA 2025
- Investigated LLM-based approaches for clinical document summarisation and classification, enabling faster information retrieval for clinicians over unstructured patient records
- Managed two data scientists, introducing git workflows, agile sprint structure, and code review practices to a team without prior engineering discipline
- Owned senior stakeholder relationships across Neurosurgery, Clinical Coding, A&E, and Operations, including board-level collaboration on health inequalities strategy
- Cloud and compute stack: Azure, AzureML, Databricks, Spark

NHS Clinical Entrepreneur, Jan 2024 – Present (2 years 5 months)

NHS England, U.K.

- Cohort 8 NHS Clinical Entrepreneur Fellow, selected to drive innovation in the NHS.
- The Clinical Entrepreneur Programme is an NHS initiative, delivered by Anglia Ruskin University. It is the biggest entrepreneurial workforce development programme of its kind.

Founding AI Engineer, June 2024 – December 2025 (1 year 6 months)

Rumii, London, U.K.

- Founding AI Engineer at Rumii (previously MindPsy), a mental health startup, to own development of the conversational components of the app, a compassionate co-pilot.
- End-to-end agentic LLM development from research and finetuning to validation, deployment and monitoring (LLMOps).
- Leading and contributing to backend architecture and system design from ground up.
- Mentoring and overseeing junior team members in a cross functional team.

OTHERS

- **Management Committee Member - NHS-R Community** (2 years 6 months)
- **EHR Development Intern - Leeds Teaching Hospitals NHS Trust** (7 months)
- **Data Analyst Intern – The Shaadi Times** (3 months)
- **Analytics Intern - Megastar Engineering** (7 months)

PROFESSIONAL SUMMARY

AI Research Engineer with **4+ years** of experience building **applied AI systems** from the ground up, across **LLM orchestration**, **RAG pipelines**, and transformer-based **NLP** in high-stakes fintech/healthtech and startup environments. Founding engineer on multiple AI products, with a track record of taking systems from **zero to production** and presenting research at national conferences. Equally at home defining technical direction in early-stage startups and navigating complexity in large organisations, with hands-on experience leading engineering teams and influencing data-driven strategy at board level.

EDUCATION

MSc. Data Science and Analytics, 2021-22

University of Leeds, U.K.

Distinction

B.E. Information Technology, 2016-2020

Savitribai Phule Pune University, India

Distinction (GPA: 8.75)

PROJECT HIGHLIGHTS

- **@ClawMe(atclawme.com)**: Founder/Building a persistent identity and discovery registry for self-hosted AI agents from the ground up, providing human-readable @handles, A2A-compatible agent cards, and a connection graph for the agentic web, using TypeScript, React, and Supabase.
- Built a custom LLM orchestration framework from scratch to power a mental health conversational AI (**Rumii**), designing the **core inference pipeline**, **prompt management**, **memory intelligence layer** and **safety-aware response** logic for a consumer-facing chat product.
- NLP/AI in assisting Clinical Coding (**NCA NHS FT**) for clinical text documents. Multi **agent + deterministic hybrid pipeline** for medical ontology automation at scale.
- LLM assisted discharge summary generation.
- **LLM Evaluation Framework/Harness** for healthcare deployment. Part of a larger strategic evaluation suite (**NCA NHS FT**).
- Sample size re-estimation in randomised clinical trials (MSc. Dissertation)
- Researched the impact of mis-specified nuisance parameters on sample size calculations in clinical trials. Demonstrated the feasibility of **adjusting allocation ratios** to meet financial or trial-specific needs while maintaining controlled error rates.

CORE SKILLS

- Python, TypeScript, SQL, C, C++
- Applied AI, NLP, Deep Learning, LLM Orchestration, RAG, Fine-tuning, LLMOps, MLOps
- PyTorch, HuggingFace Transformers, LangChain, FastAPI
- LLM Evaluation and Benchmarking, Agentic AI Systems, Prompt Engineering
- Vector Databases, Spark

REFERENCES AVAILABLE UPON REQUEST